UNIT 1 · LESSON 11-1 · BOOK P.629

Quiz · Line Them Up

Adding and Subtracting Polynomials · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 Which expression is NOT a polynomial? **2 marks**

A -17.95
B $8y^2 + 5$
C $-7a^2b - 3c$
D $7y^{-1} + 3$

2 Classify $-2 - 8x$ by the number of terms. **2 marks**

A monomial
B binomial
C trinomial
D not a polynomial

3 What is the degree of $-3y^2 + x^4y + 9x^4y^2$? **2 marks**

Answer: _____

4 Write $r^2 - 5r + 3 + 2r^3$ in standard form. **2 marks**

A $r^2 + 2r^3 - 5r + 3$
B $2r^3 + r^2 - 5r + 3$
C $2r^3 + r^2 + 5r + 3$
D $3 - 5r + r^2 + 2r^3$

5 Find the sum $(-6a^2 + 8a + 2) + (5a - 1)$. **2 marks**

A $-6a^2 + 3a + 3$
B $-6a^2 + 13a + 1$
C $-6a^4 + 13a + 1$
D $-6a^2 + 13a - 1$

6 Find the difference $(-y^2 + 7y + 6) - (-y + 4)$. **2 marks**

A $-y^2 + 8y + 10$
B $-y^2 + 8y + 2$
C $y^2 - 8y - 2$
D $-y^2 + 6y + 10$

7 Write $7 - 2x^2 + 9x^4 - x$ in standard form. Then state its degree, its leading coefficient, and its name by number of terms. **6 marks**

Total: _____ / 18

UNIT 1 · LESSON 11-2 · BOOK P.639

Quiz · Share It Out

Multiplying Polynomials by Monomials · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 Simplify $5b^2(-4b^2 - 5b - 6)$. **2 marks**

A $-20b^4 - 25b^3 + 30b^2$

B $-20b^4 - 5b^3 - 6b^2$

C $-20b^4 - 55b^2$

D $-20b^4 - 25b^3 - 30b^2$

2 Simplify $-3x(-5x - 6) - 5(-5x - 4)$. **2 marks**

A $15x^2 + 43x - 20$

B $15x^2 + 43x + 20$

C $15x^2 + 13x - 4$

D $15x^2 - 7x - 20$

3 Solve $3(4a - 9) - 3(3a - 9) = 12$. **2 marks**

Answer: _____

4 A rectangle has width $6x$ and length $4x + 4$. Write its area as a polynomial. **2 marks**

A $24x^2 + 24x$

B $24x + 24$

C $24x^2 + 4x$

D $20x + 8$

5 Simplify $3x(-x^2 - 4x + 3)$. **2 marks**

A $-3x^3 - 12x^2 + 9x$

B $-3x^3 - 12x^2 - 9x$

C $-3x^3 - 4x^2 + 3x$

D $-3x^2 - 3x$

6 Simplify $-5a(3a^2 - 8a + 7) - a(a + 7)$. **2 marks**

A $-15a^3 + 39a^2 - 28a$

B $-15a^3 + 41a^2 - 28a$

C $-15a^3 + 40a^2 - 34a + 7$

D $-15a^3 + 39a^2 - 42a$

7 Simplify $-4d(5d^2 - 12) + 7(d + 5)$. Show your steps. **6 marks**

Total: _____ / 18

UNIT 1 · LESSON 11-3 · BOOK P.645

Quiz · Every Term Meets

Multiplying Polynomials · 16 marks · 25 minutes

Name _____ Class _____ Score _____

1 Find $(4x + 7)(4x + 8)$. **2 marks**

A $16x^2 + 56$
B $16x^2 + 4x + 56$
C $16x^2 + 60x + 15$
D $16x^2 + 60x + 56$

2 Find $(k - 5)(4k + 7)$. **2 marks**

A $4k^2 - 13k - 35$
B $4k^2 + 27k - 35$
C $4k^2 - 35$
D $4k^2 - 13k + 2$

3 Find $(2x - 4)(x^2 - x + 2)$. **2 marks**

A $2x^3 - 6x^2 - 8$
B $2x^3 - 10x^2 + 8x - 8$
C $2x^3 - 2x^2 + 4x - 4$
D $2x^3 - 6x^2 + 8x - 8$

4 A rectangle is $4x + 9$ long and $x + 2$ wide. Which polynomial is its area? **2 marks**

A $4x^2 + 17x + 18$
B $10x + 22$
C $4x^2 + 18$
D $4x^2 + 18x + 18$

5 Find $(3x - 3)(3x + 6)$. **2 marks**

A $9x^2 + 9x + 3$
B $9x^2 - 18$
C $9x^2 + 27x - 18$
D $9x^2 + 9x - 18$

6 Find $(k + 2)(2k + 6)$. **2 marks**

A $2k^2 + 2k + 12$
B $2k^2 + 10k + 8$
C $2k^2 + 12$
D $2k^2 + 10k + 12$

7 Find $(2x - 2)(x^2 + 4x + 1)$. **2 marks**

A $2x^3 + 4x^2 - 6x - 2$
B $2x^3 + 6x^2 - 6x - 2$
C $2x^3 + 8x^2 + 2x - 2$
D $2x^3 + 6x^2 - 2$

8 A rectangle is $x + 1$ long and $3x + 2$ wide. Which polynomial is its area? **2 marks**

A $8x + 6$
B $3x^2 + 2$
C $3x^2 + 6x + 2$
D $3x^2 + 5x + 2$

Total: _____ / 16

UNIT 1 · LESSON 11-4 · BOOK P.653

Quiz · Shortcut Squares

Special Products · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 Find $(3x + 9)^2$. **2 marks**

A $9x^2 + 18$
B $9x^2 + 54x + 81$
C $9x^2 + 81$
D $9x^2 + 27x + 81$

2 Find $(3y + 1)^2$. **2 marks**

A $9y^2 + 6y + 1$
B $9y^2 + 2$
C $9y^2 + 3y + 1$
D $9y^2 + 1$

3 Find $(3n + 2)(3n - 2)$. **2 marks**

A $9n^2 - 4$
B $9n^2 - 12n - 4$
C $9n^2 + 4$
D $9n^2 - 5$

4 Find $(2k - 3)^2$. **2 marks**

A $4k^2 - 12k - 9$
B $4k^2 - 6k + 9$
C $4k^2 + 9$
D $4k^2 - 12k + 9$

5 Find $(k - 4)^2$. **2 marks**

A $k^2 + 16$
B $k^2 - 8k - 16$
C $k^2 - 4k + 16$
D $k^2 - 8k + 16$

6 Find $(5x + 2)^2$. **2 marks**

A $25x^2 + 10x + 4$
B $25x^2 + 4$
C $25x^2 + 20x + 5$
D $25x^2 + 20x + 4$

7 Find each product: (a) $(5y + 4)^2$ (b) $(2a - 3)^2$ (c) $(x^2 + 3y)(x^2 - 3y)$. **6 marks**

Total: _____ / 18

UNIT 1 · LESSON 11-5 · BOOK P.661

Quiz · Pull It Out

Using the Distributive Property · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 What is the GCF of $21x^3y$ and $35x^2y^2$? **2 marks**

A $14x^2$ B $7x^3y^2$

C 7 D $7x^2y$

2 Factor completely: $21t^2 - 12t$. **2 marks**

A $3t(7t - 5)$ B $3(7t^2 - 4t)$

C $3t(7t - 3)$ D $3t(7t - 4)$

3 Factor completely: $15a^3 - 6a^2$. **2 marks**

A $3a^2(5a - 2)$ B $3a(5a^2 - 2a)$

C $3(5a^3 - 2a^2)$ D $3a^2(5a - 1)$

4 Factor by grouping: $3mp + 2m - 15p - 10$. **2 marks**

A $(m + 5)(3p - 2)$ B $(m - 5)(3p + 2)$

C $(m + 2)(3p - 5)$ D $(3m - 5)(p + 2)$

5 Factor by grouping: $y^3 - 7y^2 - 7y + 49$. **2 marks**

A $(y - 7)(y^2 - 8)$ B $(y - 7)(y^2 - 7)$

C $(y + 7)(y^2 - 7)$ D $(y - 7)(y^2 + 7)$

6 What is the GCF of $4x^2$ and $12xy$? **2 marks**

A $8x$ B 4

C $4x$ D $4x^2y$

7 Factor $6x^3y - 9x^2y^2 + 12x^2y$ completely, then check by multiplying. **6 marks**

Total: _____ / 18

UNIT 1 · LESSON 11-6 · BOOK P.667

Quiz · Find the Pair

Factoring Quadratic Trinomials · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 Factor $n^2 - 12n + 32$. **2 marks**

A $(n - 12)(n + 1)$
B $(n - 8)(n + 4)$
C $(n + 8)(n + 4)$
D $(n - 8)(n - 4)$

2 Factor $y^2 - 2y - 63$. **2 marks**

A $(y - 9)(y + 7)$
B $(y + 9)(y - 7)$
C $(y + 2)(y + 1)$
D $(y - 9)(y - 7)$

3 Factor $5x^2 - 27x + 28$. **2 marks**

A $(5x - 4)(x - 7)$
B $(5x + 7)(x + 4)$
C $(5x - 7)(x + 4)$
D $(5x - 7)(x - 4)$

4 Which trinomial is **prime** (cannot be factored using integers)? **2 marks**

A $x^2 - 3x - 20$
B $x^2 + 9x + 14$
C $x^2 - 7x + 12$
D $x^2 + 6x - 16$

5 Factor $m^2 - 4m + 3$. **2 marks**

A $(m - 1)(m + 3)$
B $(m + 1)(m + 3)$
C $(m - 1)(m - 3)$
D $(m - 4)(m + 1)$

6 Factor $y^2 - 2y + 1$. **2 marks**

A $(y - 2)(y + 1)$
B $(y + 1)(y + 1)$
C $(y - 1)(y + 1)$
D $(y - 1)(y - 1)$

7 A square has area $x^2 - 6x + 9$ ft². Its length is increased by 4 ft and its width stays the same. Find the area of the new rectangle. **6 marks**

Total: _____ / 18

UNIT 1 · LESSON 11-7 · BOOK P.675

Quiz · Spot the Square

Factoring Special Products · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 Factor $49y^2 - 9$. **2 marks**

A $(7y + 3)^2$
B $(7y - 3)^2$
C $(49y + 3)(y - 3)$
D $(7y + 3)(7y - 3)$

2 Factor completely: $5x^2 - 320$. **2 marks**

A $5(x + 8)(x - 8)$
B $(x + 8)(x - 8)$
C $(5x + 8)(x - 8)$
D $5(x - 8)^2$

3 Factor the perfect square trinomial $9x^2 + 24x + 16$. **2 marks**

A $(9x + 4)^2$
B $(3x + 4)^2$
C $(3x + 4)(3x - 4)$
D $(3x - 4)^2$

4 Factor $81v^2 - 49$. **2 marks**

A $(9v - 7)^2$
B $(81v + 7)(v - 7)$
C $(9v + 7)^2$
D $(9v + 7)(9v - 7)$

5 Factor $36x^2 - 121$. **2 marks**

A $(6x + 11)(6x - 11)$
B $(6x + 11)^2$
C $(36x + 11)(x - 11)$
D $(6x - 11)^2$

6 Factor completely: $3x^2 - 243$. **2 marks**

A $(3x + 9)(x - 9)$
B $(x + 9)(x - 9)$
C $3(x + 9)(x - 9)$
D $3(x - 9)^2$

7 A square room is 6 m by 6 m. A square of side **x** m near the door is tiled; the rest is carpeted. Write the carpeted area as a polynomial, then factor it. **6 marks**

Total: _____ / 18

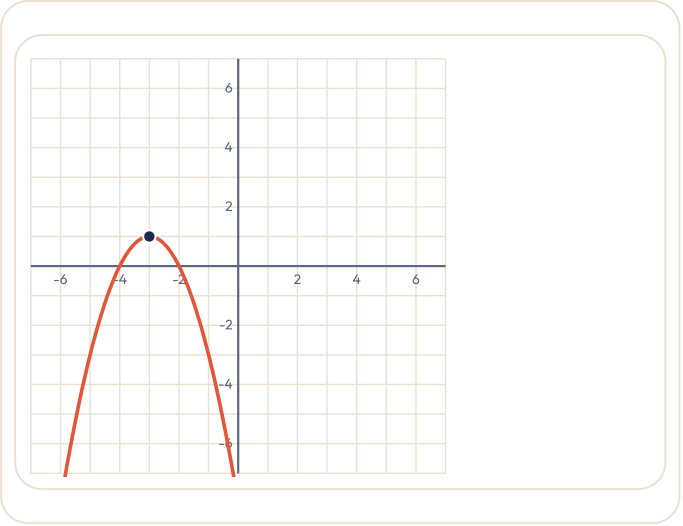
UNIT 2 · LESSON 12-1 · BOOK P.689

Quiz · The Turning Point

Graphing Quadratic Functions · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 Use the graph. What is the axis of symmetry? **2 marks**



A x = 3
B x = 1
C x = −3
D y = 1

2 What is the axis of symmetry of $y = 4x^2 - 40x + 107$? Type the number: x = ? **2 marks**

Answer: _____

3 Does $f(x) = 2x^2 - 16x + 32$ have a maximum or a minimum? What is it? **2 marks**

A minimum value 32
B minimum value 0
C maximum value 0
D minimum value 4

4 Find the vertex of $y = -x^2 - 2x - 3$. **2 marks**

A (−1, −3)
B (1, −2)
C (1, −6)
D (−1, −2)

5 A ball's height in feet after t seconds is $h(t) = -16t^2 + 48t + 40$. What is its maximum height (in feet)? **2 marks**

Answer: _____

6 A rectangle has a perimeter of 80 m, so its area is $A = x(40 - x)$. What is the greatest possible area (m^2)? **2 marks**

Answer: _____

7 A ball is thrown so that $h(t) = -16t^2 + 32t + 48$ (feet). (a) When does it reach its maximum height? (b) What is the maximum height? (c) What is its height at the start? **6 marks**

Total: _____ / 18

UNIT 2 · LESSON 12-2 · BOOK P.699

Quiz · Move the Parabola

Transformations of Quadratic Functions · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 How is the graph of $g(x) = -0.5(x + 4)^2 + 6$ related to the graph of $f(x) = x^2$? **2 marks**

A compressed vertically, translated 4 units left, translated 6 units up

B reflected across the **x**-axis, compressed vertically, translated 4 units right, translated 6 units up

C reflected across the **x**-axis, compressed vertically, translated 6 units right, translated 4 units down

D reflected across the **x**-axis, compressed vertically, translated 4 units left, translated 6 units up

2 How is the graph of $g(x) = -2(x + 2)^2$ related to the graph of $f(x) = x^2$? **2 marks**

A reflected across the **x**-axis, stretched vertically, translated 2 units left

B reflected across the **x**-axis, stretched vertically, translated 1 unit right, translated 2 units down

C stretched vertically, translated 2 units right

D reflected across the **x**-axis, compressed vertically, translated 2 units left

3 How is the graph of $g(x) = -(x + 5)^2 - 3$ related to the graph of $f(x) = x^2$? **2 marks**

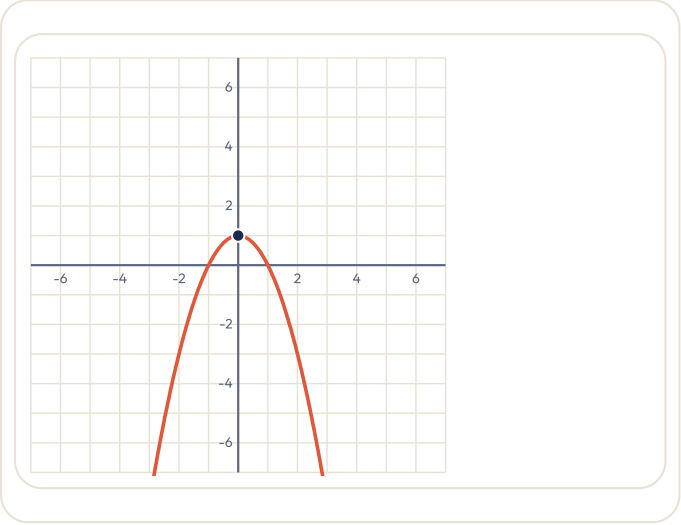
A reflected across the **x**-axis, stretched vertically, translated 5 units left, translated 3 units down

B reflected across the **x**-axis, translated 5 units left, translated 3 units down

C reflected across the **x**-axis, translated 5 units left, translated 3 units up

D reflected across the **x**-axis, translated 5 units right, translated 3 units down

4 Which equation matches the graph? **2 marks**



A $y = -x^2 + 1$

B $y = -x^2 + 2$

C $y = -x^2 - 1$

D $y = x^2 + 1$

5 What is the vertex of $g(x) = 2(x - 3)^2 - 5$? **2 marks**

A (−3, −5)

B (−5, 3)

C (3, 5)

D (3, −5)

6 How is the graph of $g(x) = 3(x + 1)^2$ related to the graph of $f(x) = x^2$? **2 marks**

A reflected across the **x**-axis, stretched vertically, translated 1 unit right

B reflected across the **x**-axis, stretched vertically, translated 1 unit left

C stretched vertically, translated 1 unit right, translated 1 unit down

D stretched vertically, translated 1 unit left

7 Describe how the graph of $g(x) = -\frac{1}{2}(x + 4)^2 - 3$ is related to $f(x) = x^2$. State its vertex and whether it has a maximum or minimum. **6 marks**

Total: _____ / 18

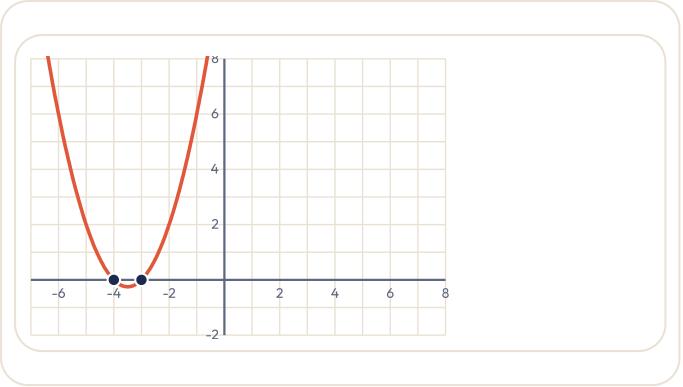
UNIT 2 · LESSON 12-3 · BOOK P.711

Quiz · Where It Lands

Solving Quadratic Equations by Graphing · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 Solve $x^2 + 7x + 12 = 0$ using the graph of the related function. **2 marks**



A $x = -4$ and $x = -3$

B $x = 12$

C $x = 4$ and $x = 3$

D $x = -3.5$

2 How many real solutions does $2x^2 + 12x + 12 = 0$ have? (Hint: the vertex of the related function is $(-3, -6)$.) **2 marks**

A no real solutions

B one real solution (a double root)

C two real solutions

D infinitely many solutions

3 How many real solutions does $x^2 - 6x + 10 = 0$ have? (Hint: the vertex of the related function is $(3, 1)$.) **2 marks**

A no real solutions

B one real solution (a double root)

C two real solutions

D infinitely many solutions

4 The greater root of $x^2 - 6x - 5 = 0$ lies between which two consecutive integers? **2 marks**

A 5 and 6

B 6 and 7

C −7 and −6

D 7 and 8

5 Estimate the greater root of $x^2 + 5x - 3 = 0$ to the nearest tenth. **2 marks**

Answer: _____

6 Estimate the greater root of $x^2 + 5x - 8 = 0$ to the nearest tenth. **2 marks**

Answer: _____

7 Solve $-x^2 + 4x + 5 = 0$ by graphing. Show the axis of symmetry, the vertex, a table and the solutions. **6 marks**

Total: _____ / 18

UNIT 2 · LESSON 12-4 · BOOK P.715

Quiz · Zero Wins

Solving Quadratic Equations by Factoring · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 Solve $(x - 6)^2 = 36$. **2 marks**

A $x = -12$ or $x = 0$

B $x = 12$ only

C $x = -30$ or $x = 42$

D $x = 0$ or $x = 12$

2 Solve $(x + 7)(x - 5) = 0$. **2 marks**

A $x = 7$ or $x = 5$

B $x = -7$

C $x = -7$ or $x = 5$

D $x = 7$ or $x = -5$

3 Solve $x^2 + 4x = 0$. **2 marks**

A $x = -4$ only

B $x = 1$ or $x = -4$

C $x = 0$ or $x = 4$

D $x = 0$ or $x = -4$

4 Solve $(x + 8)(x - 3) = 0$. **2 marks**

A $x = 8$ or $x = -3$

B $x = -8$ or $x = 3$

C $x = -8$

D $x = 8$ or $x = 3$

5 Solve $x^2 + 4x + 3 = 0$ by factoring. **2 marks**

A $x = -3$ or $x = -1$

B $x = 3$

C $x = -1$ or $x = 3$

D $x = 1$ or $x = 3$

6 Solve $64x^2 = 9$. **2 marks**

A no real solutions

B $x = \frac{3}{8}$ only

C $x = \frac{9}{64}$ or $x = -\frac{9}{64}$

D $x = \frac{3}{8}$ or $x = -\frac{3}{8}$

7 Solve $3b(9b - 27) = 0$ and $a^2 = 4a$. Check each root. **6 marks**

Total: _____ / 18

UNIT 2 · LESSON 12-5 · BOOK P.725

Quiz · Make It Square

Solving Quadratic Equations by Completing the Square · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 Find the value of **c** that makes $x^2 + 18x + c$ a perfect square trinomial. **2 marks**

A 18
B 9
C 81
D 324

2 Solve $x^2 + 6x = 40$ by completing the square. **2 marks**

A $x = 10$ or $x = -4$
B $x = -52$ or $x = 46$
C $x = -3$
D $x = -10$ or $x = 4$

3 Solve $x^2 + 6x = -5$ by completing the square. **2 marks**

A $x = -3$
B $x = -5$ or $x = -1$
C $x = 5$ or $x = 1$
D $x = -7$ or $x = 1$

4 Write $y = -x^2 - 6x - 8$ in vertex form. **2 marks**

A $y = -(x + 3)^2 - 1$
B $y = -(x + 3)^2 + 1$
C $y = -(x + 3)^2 - 8$
D $y = -(x - 3)^2 + 1$

5 Write $y = x^2 + 12x + 27$ in vertex form. **2 marks**

A $y = (x + 6)^2 + 27$
B $y = (x + 6)^2 - 9$
C $y = (x - 6)^2 - 9$
D $y = (x + 6)^2 + 9$

6 Write $y = x^2 + 6x + 1$ in vertex form. **2 marks**

A $y = (x + 3)^2 + 8$
B $y = (x + 3)^2 - 8$
C $y = (x - 3)^2 - 8$
D $y = (x + 3)^2 + 1$

7 Solve $x^2 + 6x - 16 = 0$ by completing the square. Show every step. **6 marks**

Total: _____ / 18

UNIT 2 · LESSON 12-6 · BOOK P.731

Quiz · One Formula Rules

Solving Quadratic Equations by Using the Quadratic Formula · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 Solve $2x^2 + 23x + 56 = 0$ using the Quadratic Formula. **2 marks**

A $x = -7$ or $x = -16$

B $x = -\frac{7}{2}$ or $x = -8$

C $x = \frac{7}{2}$ or $x = 8$

D $x = -\frac{7}{2}$ only

2 Solve $3x^2 + 28x + 32 = 0$ using the Quadratic Formula. **2 marks**

A $x = -\frac{4}{3}$ or $x = -8$

B $x = -\frac{8}{3}$ or $x = -16$

C $x = \frac{4}{3}$ or $x = 8$

D $x = -\frac{4}{3}$ only

3 Solve $x^2 - 6x - 7 = 0$ using the Quadratic Formula. **2 marks**

A $x = 14$ or $x = -2$

B $x = -7$ or $x = 1$

C $x = 7$ only

D $x = 7$ or $x = -1$

4 Use the discriminant to describe the roots of $-2x^2 - 4x - 9 = 0$. **2 marks**

A two real rational roots

B two real irrational roots

C one real rational root

D no real roots

5 Solve $x^2 - 8x - 6 = 0$. Give exact answers. **2 marks**

A $x = 4 \pm \sqrt{22}$

B $x = 5 \pm \sqrt{22}$

C $x = 4 \pm \sqrt{88}$

D $x = -4 \pm \sqrt{22}$

6 Solve $2x^2 + x - 21 = 0$ using the Quadratic Formula. **2 marks**

A $x = -\frac{7}{2}$ or $x = 3$

B $x = \frac{7}{2}$ or $x = -3$

C $x = -\frac{7}{2}$ only

D $x = -7$ or $x = 6$

7 $P = 0.01t^2 + 0.5t + 107$ models blood pressure at age t. Find the age when $P = 120$. Round to the nearest tenth. **6 marks**

Total: _____ / 18

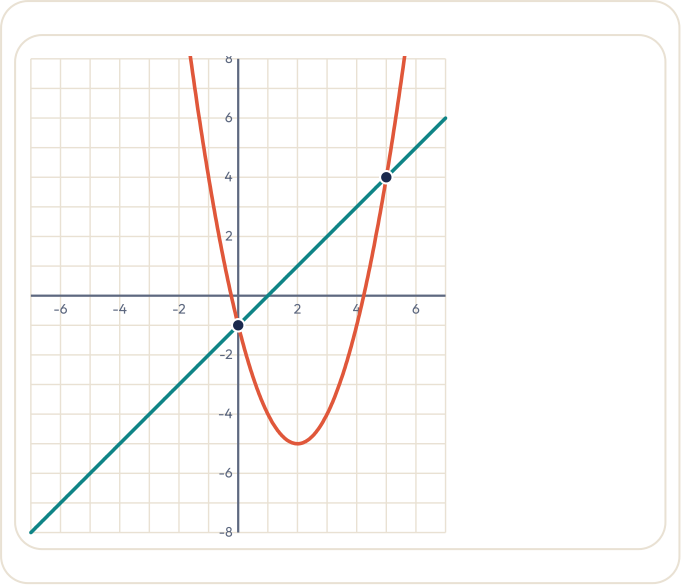
UNIT 2 · LESSON 12-7 · BOOK P.739

Quiz · Where They Meet

Solving Systems of Linear and Quadratic Equations · 16 marks · 25 minutes

Name _____ Class _____ Score _____

1 Use the graph to solve the system of the parabola and the line. **2 marks**



A **(0, −1)**
B **(−1, 0)** and **(4, 5)**
C **(0, −1)** and **(5, 4)**
D **(0, 0)** and **(5, 0)**

2 Solve the system: $y = x^2 - 5x - 5$ and $y = -3x - 5$. **2 marks**

A **(0, −5)** and **(2, −11)**
B **(−5, 0)** and **(−11, 2)**
C **(0, −11)** and **(2, −5)**
D **(0, −5)** and **(−2, −11)**

3 How many solutions does the system $y = x^2 - 4x + 5$, $y = -x - 2$ have? **2 marks**

A no solution
B one solution
C two solutions
D infinitely many solutions

4 Solve the system: $y = x^2 + 2x + 4$ and $y = 3x + 4$. **2 marks**

A **(0, 4)** and **(−1, 7)**
B **(4, 0)** and **(7, 1)**
C **(0, 7)** and **(1, 4)**
D **(0, 4)** and **(1, 7)**

5 Solve the system: $y = x^2 - 7x - 1$ and $y = -3x - 4$. **2 marks**

A **(−1, −7)** and **(−3, −13)**
B **(−7, 1)** and **(−13, 3)**
C **(1, −13)** and **(3, −7)**
D **(1, −7)** and **(3, −13)**

6 How many solutions does the system $y = x^2 + 3x + 1$, $y = -x - 4$ have? **2 marks**

A no solution
B one solution
C two solutions
D infinitely many solutions

7 Solve the system: $y = x^2 + 9x + 10$ and $y = 3x + 2$. **2 marks**

A $(-4, -10)$ and $(-2, -4)$

B $(-10, -4)$ and $(-4, -2)$

C $(-4, -4)$ and $(-2, -10)$

D $(4, -10)$ and $(2, -4)$

8 Solve the system: $y = x^2 - 10$ and $y = 3x$. **2 marks**

A $(-6, -2)$ and $(15, 5)$

B $(2, -6)$ and $(-5, 15)$

C $(-2, -6)$ and $(5, 15)$

D $(-2, 15)$ and $(5, -6)$

Total: _____ / 16

UNIT 3 · LESSON 10-1 · BOOK P.571

Quiz · Power Up

Multiplication Properties of Exponents · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 Which expression is NOT a monomial? **2 marks**

A x^{-1}

B $\frac{5x^3}{4}$

C -17

D a^3

2 Simplify $(-4a)(-a^4b)$. **2 marks**

A $4a^5b$

B $-5a^5b$

C $4a^4$

D $-4a^5b$

3 Simplify $(-6mn^2)(5m^3n^2)$. **2 marks**

A $-30m^3n^4$

B $-m^4n^4$

C $-30m^4n^4$

D $30m^4n^4$

4 Simplify $(n^4)^2 \cdot n^6 = n^?$. Type the exponent. **2 marks**

Answer: _____

5 Simplify $(-2ax^3)^2$. **2 marks**

A $-2a^2x^6$

B $4a^3x^5$

C $-4a^2x^6$

D $4a^2x^6$

6 Simplify $(5xy^4)^4$. **2 marks**

A $625x^4y^{16}$

B $5x^4y^{16}$

C $625x^5y^8$

D $20x^4y^{16}$

7 Simplify $(11x^6y^8)(xy^9)$ and $(-4m^3n)(-2mn^5)$. **6 marks**

Total: _____ / 18

UNIT 3 · LESSON 10-2 · BOOK P.579

Quiz · Subtract the Powers

Division Properties of Exponents · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 Simplify $\frac{-15n^7y^4}{5n^2y^2}$. Assume no denominator is 0. **2 marks**

A $-20n^5y^2$
B $3n^5y^2$
C $-3n^4y^2$
D $-3n^5y^2$

2 Simplify $(\frac{4p^3}{7})^2$. **2 marks**

A $\frac{16p^6}{49}$
B $\frac{16p^6}{7}$
C $\frac{4p^6}{49}$
D $\frac{16p^5}{49}$

3 Simplify $\frac{-10n^3x^4}{-2n^2x^2}$. Assume no denominator is 0. **2 marks**

A $-5nx^2$
B $-8nx^2$
C $5nx^2$
D $5n^2x^2$

4 Simplify $\frac{8b^3y^4}{-4by^2}$. Assume no denominator is 0. **2 marks**

A $-2b^3y^2$
B $2b^2y^2$
C $12b^2y^2$
D $-2b^2y^2$

5 Simplify $(\frac{5y^4}{2})^3$. **2 marks**

A $\frac{125y^7}{8}$
B $\frac{5y^{12}}{8}$
C $\frac{125y^{12}}{2}$
D $\frac{125y^{12}}{8}$

6 Simplify $\frac{3a^4x^7}{-a^3x^3}$. Assume no denominator is 0. **2 marks**

A $3ax^4$
B $-3ax^2$
C $-3ax^4$
D $4ax^4$

7 Simplify $\frac{a^2b^5ca^4}{b^2cd}$ and $(\frac{4m^2n^2p^2}{3mp})^4$. Assume no denominator is 0. **6 marks**

Total: _____ / 18

UNIT 3 · LESSON 10-3 · BOOK P.587

Quiz · Flip It Over

Negative Exponents · 18 marks · 25 minutes

Name _____ Class _____ Score _____

1 Simplify $\frac{4a^2b^0}{a^2}$. Assume no denominator is 0. **2 marks**

A -1
B 4
C 1
D 0

2 Simplify $3m^{-4}x^2$ so it has only positive exponents. **2 marks**

A $-3m^4x^2$
B $\frac{3x^2}{m^4}$
C $\frac{x^2}{3m^4}$
D $3m^4x^2$

3 Simplify $(-6xy)^0$. Assume no denominator is 0. **2 marks**

A -1
B not possible
C 0
D 1

4 Simplify $\frac{7a^5b^0}{a^2}$. Assume no denominator is 0. **2 marks**

A -1
B 0
C $7a^3$
D 1

5 Simplify $4b^{-3}x^4$ so it has only positive exponents. **2 marks**

A $-4b^3x^4$
B $\frac{x^4}{4b^3}$
C $\frac{4x^4}{b^3}$
D $4b^3x^4$

6 Simplify $6b^{-5}x^3$ so it has only positive exponents. **2 marks**

A $\frac{6x^3}{b^5}$
B $6b^5x^3$
C $-6b^5x^3$
D $\frac{x^3}{6b^5}$

7 Simplify each with positive exponents: (a) $\frac{3g^{-3}h^2}{-36g^3h^{-4}}$ (b) $\frac{35r^3s^{-1}}{14r^4s^2}$. **6 marks**

Total: _____ / 18